Laboratory 1a

Raw Security Logs to Rule-Based Detection, MITRE ATT&CK, and SOC Dashboard

Golan Katzav and Lior Masiach | October 25, 2026

Execution record. C1-C4 were run in the Docker Compose lab with the optional simulator stopped. After each log edit, we waited for the detector scan and compared the dashboard with output/findings.json. The observed results matched the predictions recorded below.

1. Clean baseline and execution record

The six-line baseline contains three failed-password records from 172.18.0.5 and one explicit network-scan record from 172.18.0.7. The original report records that the dashboard displayed two findings: one High-severity T1110 finding and one Medium-severity T1046 finding.

Check	Recorded result	Interpretation
Initial finding count	2: T1110 from 172.18.0.5; T1046 from 172.18.0.7.	Matches the clean-baseline result described in the handout and original report.
Docker / dashboard verification	Docker Compose detector and dashboard were running. The dashboard opened and displayed the clean-baseline count of two. During C1-C4, the dashboard and output/findings.json were checked after each detector scan.	The runtime remained available during the controlled experiments; the optional simulator was stopped while C1-C4 were run.

2. Evidence-to-rule mapping

Raw evidence	Rule and finding	ATT&CK technique and tactic	Uncertainty
Three baseline lines say "Failed password" and identify source 172.18.0.5 (ports 44231-44233).	Count matching lines by source IP. At least three from one IP produce one High-severity candidate finding.	T1110 - Brute Force Credential Access	Shows repeated authentication failures only. It does not show a successful login, malicious intent, attacker identity, or compromise. The rule has no time window.
One baseline line explicitly says "Network scan detected from 172.18.0.7 against ports 22,80,443,3306."	A line containing the phrase "network scan detected" produces one Medium-severity candidate finding.	T1046 - Network Service Discovery Discovery	The rule matches text in the log; it does not infer scanning from packet or connection-flow evidence. The record alone does not establish intent or confirm a real scan.

3. Activity C - controlled experiments

C1-C4 were executed sequentially from the clean baseline with the simulator stopped. After each edit, the detector was allowed to rescan and the dashboard and output/findings.json were checked. The observed results below matched the predictions.

Experiment and changed log data	Prediction	Observed result	Why
C1 - Remove one failure Remove the third failed-password line: 2026-07-11T09:02:18 WARN Failed password for root from 172.18.0.5 port 44233 ssh2	T1110 disappears; baseline T1046 remains; total = 1.	Observed in the dashboard and output/findings.json: T1110 disappeared; T1046 from 172.18.0.7 remained. Total = 1; prediction matched.	Only two failures remain for 172.18.0.5, below the threshold of three.
C2 - Distribute three failures Restore three lines, with one failure per source IP: 2026-07-11T09:02:11 WARN Failed password for root from 172.18.0.5 port 44231 ssh2 2026-07-11T09:02:14 WARN Failed password for root from 172.18.0.6 port 44232 ssh2 2026-07-11T09:02:18 WARN Failed password for root from 172.18.0.8 port 44233 ssh2	No T1110; baseline T1046 remains; total = 1.	Observed in the dashboard and output/findings.json: no T1110 appeared; only T1046 remained. Total = 1; prediction matched.	The rule groups by source IP; three failures spread across three IPs do not combine into one threshold count.
C3 - Cross the threshold from one new IP Append: 2026-10-22T10:01:00Z WARN Failed password for test from 192.0.2.55 port 40001 ssh2 2026-10-22T10:01:02Z WARN Failed password for test from 192.0.2.55 port 40002 ssh2 2026-10-22T10:01:04Z WARN Failed password for test from 192.0.2.55 port 40003 ssh2	Add T1110 from 192.0.2.55; T1046 remains; total = 2.	Observed in the dashboard and output/findings.json: a new High T1110 appeared for 192.0.2.55, alongside the baseline T1046. Total = 2; prediction matched.	Three new failures from the same source meet the threshold. The documentation-only address keeps the test synthetic.
C4 - Add a second scan record Append: 2026-10-22T10:02:00Z INFO Network scan detected from 192.0.2.77 against ports 22,80,443	Add another T1046 from 192.0.2.77; total = 3.	Observed in the dashboard and output/findings.json: one T1110 from 192.0.2.55 and two T1046 findings (172.18.0.7, 192.0.2.77). Total = 3; prediction matched.	Each explicit scan-marker line creates a T1046 finding; this rule does not validate network traffic independently.

Final count before the simulator

The observed final finding count immediately after C4 and before starting the simulator was 3 findings: one High T1110 from 192.0.2.55 and two Medium T1046 findings from 172.18.0.7 and 192.0.2.77. It matched the predicted output.

4. Activity D - simulator observation

The original report records that the simulator introduced source IPs 172.18.0.22 and 172.18.0.33. Repeated failed-password events from 172.18.0.22 matched T1110, and an explicit network-scan notice from 172.18.0.33 matched T1046. The reported finding count increased as synthetic events accumulated. An exact final simulator count was not recorded. Because events are generated over time, the simulator is less reproducible than the fixed C1-C4 log edits.

5. SOC narrative

The clean six-line baseline was run in Docker Compose, and the dashboard showed two findings: High T1110 from 172.18.0.5 and Medium T1046 from 172.18.0.7. We then ran C1-C4 with the simulator stopped, waiting for a detector scan after each log edit and checking both the dashboard and output/findings.json. C1 left only the baseline T1046 (one finding). C2 distributed three failed logins over three IPs and still left one finding. C3 produced a new T1110 from 192.0.2.55, bringing the total to two. C4 added T1046 from 192.0.2.77, bringing the pre-simulator total to three. The simulator later produced events from 172.18.0.22 and 172.18.0.33 and the count increased. The alerts are rule matches, not confirmed attacks: the brute-force rule has no time window, the scan rule depends on explicit log wording, and neither proves intent, successful access, attribution, or compromise.

Word count: 137.

References

1. Lab 1a student handout and project (https://github.com/melofon/HIT-ai-cybersecurity-labs/tree/main/labs/lab1a-mitre-soc).
2. Supplied detector source (https://raw.githubusercontent.com/melofon/HIT-ai-cybersecurity-labs/main/labs/lab1a-mitre-soc/detector.py).
3. MITRE ATT&CK: T1110 Brute Force (https://attack.mitre.org/techniques/T1110/).
4. MITRE ATT&CK: T1046 Network Service Discovery (https://attack.mitre.org/techniques/T1046/).